

ROXSTAR

AI VOICE ROOM ASSISTANT

Candidate Technical Assignment

Objective

Build two AI participants that join a LiveKit voice room, understand multi-user voice and text conversations, maintain session context, and respond naturally in Hindi/Hinglish while also understanding English.

Real-time Voice AI | LiveKit | Hindi/Hinglish | Multi-user Context

Confidential candidate assessment

Please do not publish the assignment or submitted solution.

1. Assignment Overview

Create a working AI-only prototype for a LiveKit voice chat room. Human users and AI bots must communicate in real time through voice and text. This assessment evaluates AI orchestration, speech processing, natural Indian conversation, context management and real-time engineering.

Expected journey

- A user creates or joins a LiveKit room.
- At least two human users join the same room.
- Two AI bots join as visible participants: one male voice and one female voice.
- The bots listen to the discussion and decide when a response is relevant.
- Users communicate through live voice or room text chat.
- The selected bot responds through voice and/or text while preserving context.

2. Mandatory Requirements

2.1 LiveKit room

- Use LiveKit for real-time room and audio transport.
- Support participant join, leave and reconnect.
- AI bots must appear as identifiable room participants.
- Demonstrate with at least two human participants.

2.2 Two AI bots

Bot	Voice	Expectation
Roxstar AI Dost	Male	Friendly Hindi/Hinglish persona with a natural Indian Hindi accent.
Roxstar AI Sathi	Female	Distinct Hindi/Hinglish persona with a natural Indian Hindi accent.

The solution must prevent both bots from talking simultaneously without a clear reason. Implement and document a response-routing or turn-selection strategy.

2.3 Voice understanding and participation

- Convert room speech into text or semantic input.
- Associate utterances with the correct participant where technically available.
- Handle pauses, incomplete sentences, Hinglish and normal transcription errors.
- Do not answer every sentence. Respond when addressed or contextually appropriate.
- Support questions from voice and text chat using shared room context.

2.4 Context and interruptions

- Maintain multi-turn context during the active room session.
- Resolve follow-ups such as "uski", "wahi topic" and "simple batao".
- Keep speaker-specific facts separate where possible.
- Support barge-in: stop or cancel a response when the user interrupts.
- Document context-window, summarization or truncation strategy.

3. Mandatory Language Personality

Indian conversational Hindi/Hinglish

The bots must sound like natural participants in an Indian voice room. Use everyday Hindi/Hinglish, not pure, textbook, literary or literal translation-style Hindi. English words may be mixed naturally.

3.1 Required behavior

- Understand spoken and typed Hindi.
- Understand spoken and typed Hinglish in Roman script.
- Understand English questions and instructions.
- Reply mainly in Hindi/Hinglish unless the user requests English.
- Follow the user's language style where practical.
- Use a clearly Indian Hindi-speaking accent for voice replies.

3.2 Natural language benchmark

Avoid formal / unnatural	Prefer conversational Hindi/Hinglish
AI ek aisi takneek hai jo machines ko nirnay lene mein saksham banati hai.	AI ek aisi technology hai jo machine ko samajhne aur decision lene layak banati hai.
Kripya pratiksha karein.	Ek second ruk jao.
Kaksh mein pravesh karein.	Room join karo.
Dhwanigrahak sakriya karein.	Mic unmute karo.

These examples define the expected overall style, not a fixed banned-word list.

3.3 Voice quality

- Male and female voices must be clearly distinguishable.
- Mixed words such as technology, decision, room, mic and network should sound natural inside Hindi sentences.
- Robotic pacing, incorrect pauses or foreign-sounding Hindi pronunciation will reduce the score.
- Document TTS provider, voice selection and configuration.

4. Required Conversation Scenarios

Scenarios may begin through voice or text chat. Demonstrate at least Scenarios 1-5.

Scenario 1: Natural Hinglish

User: AI kya hota hai?
 Bot: AI ek aisi technology hai jo machine ko samajhne aur decision lene layak banati hai.
 Simple words mein bolo to machine ko thoda smart bana deti hai.

Scenario 2: English understanding

User: Can you explain cloud computing?
 Bot: Bilkul. Cloud computing ka simple matlab hai internet ke through servers, storage aur computing resources use karna, bina sab kuch khud manage kiye.

Scenario 3: Follow-up context

User: Shah Rukh Khan ke baare mein batao.
 Bot: Shah Rukh Khan India ke sabse popular actors mein se ek hain.
 User: Unki koi famous movie batao.
 Bot: Unki famous movies mein Dilwale Dulhania Le Jayenge aur Chak De! India shamil hain.

Scenario 4: Multi-user context

Rahul: AI kya hota hai?
 Bot: AI machine ko data se seekhne aur smart decisions lene mein help karta hai.
 Priya: Thoda aur simple batao.
 Bot: Simple bolun to AI machine ka smart dimaag jaisa hai, jo examples se seekhta hai.

Scenario 5: Session memory

Rahul: Mera naam Rahul hai aur mujhe cricket pasand hai.
 Later, Rahul: Maine apne baare mein kya bataya tha?
 Bot: Tumne bataya tha ki tumhara naam Rahul hai aur tumhe cricket pasand hai.

Scenario 6: Interruption

Bot: Machine learning mein pehle data collect karte hain, phir...
 User: Ruko, simple example se samjhao.
 Bot: Theek hai. Jaise phone photos dekh kar faces pehchanna seekhta hai, waise hi...

Scenario 7: Two-bot routing

User: AI Dost, tum answer karo. AI Sathi, baad mein ek example dena.
 AI Dost: Theek hai, pehle main short answer deta hoon...
 AI Sathi: Aur iska ek simple example ye hai...

5. Technical Expectations

Suggested pipeline

LiveKit audio / text -> Speaker identification -> STT and language detection
 -> Turn detection and bot routing -> Room context and memory
 -> LLM with Hindi/Hinglish persona -> Hindi-accented TTS
 -> LiveKit audio publication and/or room text reply

Technology flexibility

Candidates may choose any suitable STT, LLM and TTS providers. LiveKit is the expected room layer. Document external services, models, configuration, limitations and cost considerations. Do not commit secrets.

Reliability, latency and privacy

- Avoid duplicate replies and uncontrolled overlapping bot speech.
- Handle temporary STT, LLM, TTS and network failures without crashing.
- Log transcription, selected bot, model response and publication stages without exposing secrets.
- Measure and report observed end-to-end response latency.
- Show that AI participants are present.
- Do not use real customer data or confidential recordings.
- Do not store raw room audio by default; disclose and document deletion if storage is used.

6. Optional Bonus Features

- Conversation summary: answer "Abhi tak kya discuss hua?"
- Improved speaker identification and speaker-specific memory.
- Basic moderation for toxic or abusive speech.
- Dynamic bot selection based on name, language, persona or topic.
- Automated tests for natural Hindi/Hinglish style and relevance.

7. Deliverables

- Private Git repository with complete source code.
- README with prerequisites, environment variables and clean-start instructions.
- Architecture document with component and sequence diagrams.
- STT, LLM, TTS and LiveKit technology decisions.
- Prompt/persona configuration for both bots.
- Documented bot-routing strategy.
- Tests for context, language, routing and failures.
- A 5-10 minute recorded demonstration.
- Known limitations, cost considerations and next steps.

8. Demonstration Checklist

- LiveKit room with at least two human users.
- Both AI bots shown as participants.
- Hindi/Hinglish voice question and Hindi-accented voice reply.
- English question and natural Hindi/Hinglish reply.
- Text-chat question and context-aware reply.
- Multi-turn follow-up and multi-user context.
- Bot routing without unnecessary double replies.
- Interruption of an in-progress bot response.
- One expected provider or network failure.
- Logs or metrics for one end-to-end request.

9. Evaluation Rubric - 100 Points

Area	Points	Strong evidence
LiveKit and real-time integration	15	Reliable room audio/text, lifecycle and reconnection.
Voice AI pipeline	15	Clear STT-LLM-TTS flow, interruption and failure handling.
Hindi/Hinglish conversation quality	20	Natural everyday code-mixing; avoids formal translation-style Hindi.
Hindi-accented voice quality	10	Indian Hindi accent, natural pronunciation, pacing and distinct voices.
Context and multi-user awareness	15	Accurate follow-ups, room memory and speaker context.
Two-bot orchestration	10	Distinct personas and controlled routing.
Architecture and code quality	10	Modular design, tests, observability, security and maintainability.

Area	Points	Strong evidence
Documentation and demo	5	Fast setup, accurate diagrams, limitations and convincing evidence.

Quality gate

A basic LLM-to-LiveKit connection is not enough. Formal, unnatural Hindi, poor accent quality or uncontrolled bot interruptions should significantly reduce the score.

10. Submission Checklist

- ☐ Private repository access
- ☐ README and setup instructions
- ☐ Male and female bots
- ☐ Hindi/Hinglish and English understanding
- ☐ Natural conversational style and Hindi accent
- ☐ Voice and text interaction
- ☐ Context, interruption and routing
- ☐ Architecture and sequence diagrams
- ☐ Tests and failure handling
- ☐ Demo video
- ☐ No secrets or personal data